

9.3 Resistivity

Question Paper

Course	CIE A Level Physics
Section	9. Electricity
Topic	9.3 Resistivity
Difficulty	Easy

Time allowed: 20

Score: /10

Percentage: /100

Question 1

The unit of resistivity, expressed in terms of base units, is given by

$$kg \, x^3 \, y^{-2} \, z^{-3}$$

Which base units are x , y and z ?

	x	y	z
A	ampere	metre	second
B	metre	ampere	second
C	metre	second	ampere
D	second	ampere	metre

[1 mark]

Question 2

A cylindrical wire of length 10 m and diameter 2.0 mm has a resistance of $0.050 \, \Omega$.

From which material is the wire made?

	Material	Resistivity / $\Omega \text{ m}$
A	bronze	1.6×10^{-7}
B	nichrome	1.6×10^{-6}
C	silver	1.6×10^{-8}
D	zinc	6.3×10^{-8}

[1 mark]

Question 3

The resistance of one device R_1 is designed to change with temperature; the resistance of another device R_2 is designed to change with illumination.

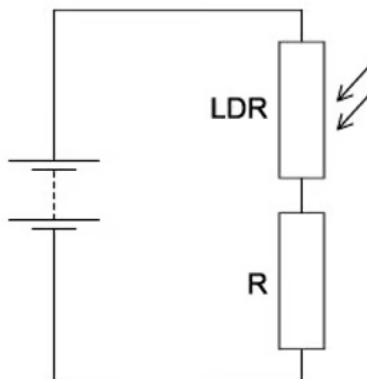
What are the names of these devices ?

	R_1	R_2
A	thermistor	light-dependent resistor
B	thermistor	potential divider
C	semiconductor diode	light-dependent resistor
D	semiconductor diode	potential divider

[1 mark]

Question 4

A light-dependent resistor (LDR) is connected in series with a resistor R and a battery.



The resistance of the LDR is equal to the resistance of R when no light falls on the LDR. When the light intensity falling on the LDR increases, which statement is correct?

- A. the current in R decreases
- B. the current through the LDR decreases
- C. the p.d. across R decreases
- D. the p.d. across the LDR decreases

[1 mark]

Question 5

A copper wire is stretched so that its diameter is reduced from 1.0 mm to a uniform 0.5 mm. The resistance of the unstretched copper wire is $0.2\ \Omega$.

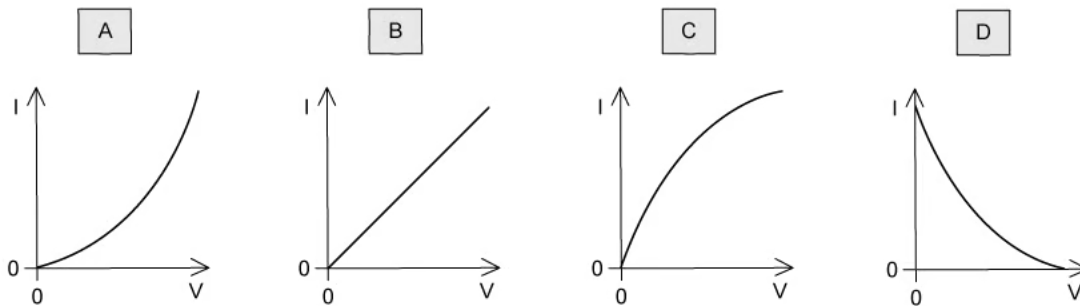
What is the resistance of the stretched wire?

- A. $0.4\ \Omega$
- B. $0.8\ \Omega$
- C. $1.6\ \Omega$
- D. $3.2\ \Omega$

[1 mark]

Question 6

Which graph best represents the way in which the current I through a thermistor depends upon the potential difference V across it?



[1 mark]

Question 7

A power cable has length 2000 m. The cable is made of twelve parallel strands of copper wire, each with diameter 0.51 mm.

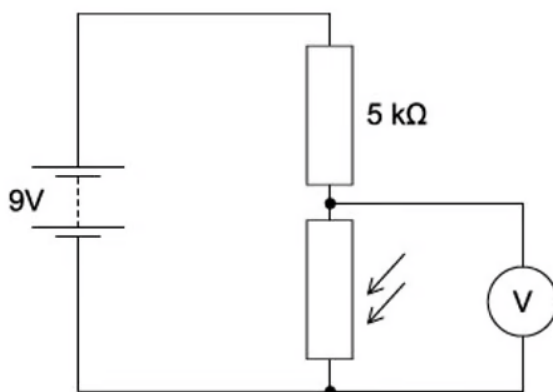
What is the resistance of the cable? (resistivity of copper = $1.7 \times 10^{-8}\ \Omega\text{ m}$)

- A. $0.014\ \Omega$
- B. $3.5\ \Omega$
- C. $14\ \Omega$
- D. $166\ \Omega$

[1 mark]

Question 8

A circuit is set up with an LDR and a fixed resistor as shown.



The voltmeter reads 4 V.

The light intensity is increased.

What is a possible voltmeter reading?

- A. 3V
- B. 4V
- C. 6V
- D. 8V

[1 mark]

Question 9

A cylindrical wire 4.0 m long has a resistance of $31\ \Omega$ and is made of metal of resistivity $1.0 \times 10^{-6}\ \Omega\text{ m}$.

What is the radius of cross-section of the wire?

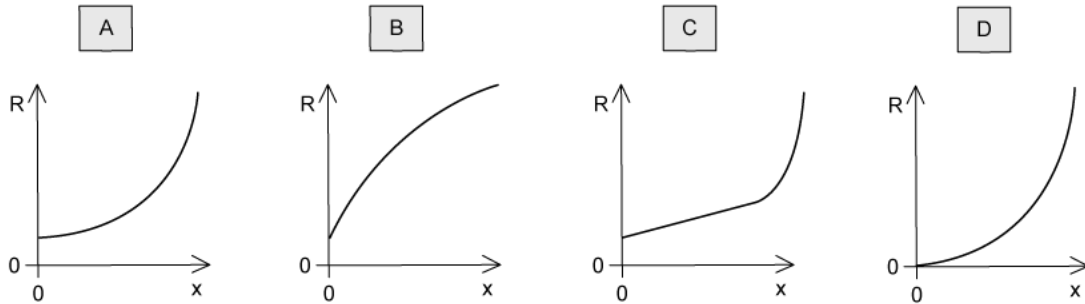
- A. $1.0 \times 10^{-8}\text{ m}$
- B. $2.0 \times 10^{-8}\text{ m}$
- C. $6.4 \times 10^{-8}\text{ m}$
- D. $2.0 \times 10^{-4}\text{ m}$

[1 mark]

Question 10

When a thin metal wire is stretched, it becomes longer and thinner. This causes a change in the resistance of the wire. The volume of the wire remains constant.

Which graph could represent the variation with extension x of the resistance R of the wire?



[1 mark]

Model Answers: Easy

1

The correct answer is **B** because:

- Resistance and resistivity are related by the equation $R = \frac{\rho L}{A}$
- So, resistivity is equal to $\rho = \frac{RA}{L}$
- Ohm's law $V = IR$ and electrical power $P = VI$ can combine to make
 - $P = (IR)I = I^2 R$
- So, resistance is equal to $R = \frac{P}{I^2}$
- Power is equal to $\frac{\text{energy}}{\text{time}}$
 - So, the units of power are $P = \frac{\frac{1}{2}mv^2}{t} = [kg \ m^2 \ s^{-2}][s]^{-1} = kg \ m^2 \ s^{-3}$
 - Therefore, the units of resistance are $R = \frac{P}{I^2} = [kg \ m^2 \ s^{-3}][A]^{-2} = kg \ m^2 \ s^{-3} \ A^{-2}$
- And hence, the units of resistivity are
 - $\rho = \frac{RA}{L} = [kg \ m^2 \ s^{-3} \ A^{-2}][m^2][m]^{-1} = kg \ m^3 \ s^{-3} \ A^{-2}$
- If we compare this to $kg \ x^3 y^{-2} z^{-3}$, we can see **x is meters, y is amperes, and z is seconds**

2

The correct answer is **C** because:

- Resistance and resistivity are related by the equation $R = \frac{\rho L}{A}$
- So, resistivity is equal to $\rho = \frac{RA}{L}$
- The wire is cylindrical, so the cross-sectional area is equal to
 - $A = \pi r^2 = \pi \left(\frac{d}{2}\right)^2 = \pi \left(\frac{2.0 \times 10^{-3}}{2}\right)^2$
- Therefore, the resistivity is equal to

$$\rho = \frac{0.050}{10} \times \pi \left(\frac{2.0 \times 10^{-3}}{2} \right)^2 = 1.57 \times 10^{-8} \Omega \text{ m}$$

- Therefore, the wire is made from **silver**, which has a resistivity of $6 \times 10^{-8} \Omega \text{ m}$

3

The correct answer is **A** because:

- A device whose resistance depends on temperature is called a **thermistor**
 - For a thermistor with a negative temperature coefficient, the resistance decreases significantly with a rise in temperature
- A device whose resistance depends on illumination is called a **light-dependent resistor or LDR**
 - The higher the intensity of light on the LDR, the greater the number of electrons in the semiconducting material that can move freely. Hence, as the light intensity increases, the resistance of the LDR decrease

4

The correct answer is **D** because:

- The resistance of the LDR decreases as the light intensity increases
- Since the LDR and resistor R are connected in series, the current in both of them is the same
- Ohm's law states $V = IR$
- Since the resistance of the LDR decreases, the potential difference V across it also decreases

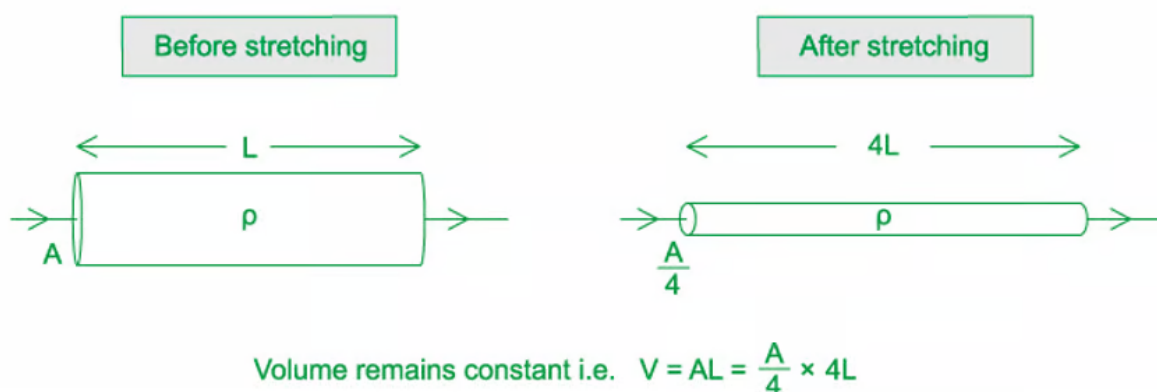
A & B are incorrect because a decrease in the resistance of the LDR causes the total resistance R in the circuit to decrease, from Ohm's law $I = \frac{V}{R}$, we can see the current through both the LDR and resistor R increases

C is incorrect because the battery has a fixed e. m. f., so this causes the p. d. across R to increase

5

The correct answer is **D** because:

- Resistance R of a wire is equal to $R = \frac{\rho L}{A}$
- Cross-sectional area is equal to $A = \pi r^2 = \pi \left(\frac{d}{2}\right)^2 = \frac{\pi d^2}{4}$
- The resistance R is inversely proportional to the area, and therefore, R is also inversely proportional to the diameter squared $R \propto \frac{1}{d^2}$
 - So, when the wire is stretched, the diameter is reduced by half, this causes the area to be reduced by 4
 - So, the resistance R is increased by a factor of 4 due to the cross-sectional area A
- Since the volume of the wire is always constant, we can use this to find the new length – the volume of the unstretched wire is $V = AL$
 - Where L is the length of the wire
- The stretched wire has $\frac{1}{4}$ the area of the unstretched wire, for its volume to remain constant, its length must have been increased by a factor of 4
 - The volume of the stretched wire is $V = \left(\frac{A}{4}\right)(4L)$
- So, the length of the stretched wire is $4L$
 - Resistance R of unstretched wire $= \frac{\rho L}{A} = 0.2 \Omega$
 - Resistance R of stretched wire $= \frac{\rho(4L)}{\frac{1}{4}A} = 16\left(\frac{\rho L}{A}\right) = 16(0.2) = 3.2 \Omega$



6

The correct answer is **A** because:

- As the potential difference V across the thermistor increases, the temperature of the thermistor also increases, causing its resistance R to decrease
 - Ohm's law states $V = IR$
- As the p.d. increases, the current increases - this is represented by a positive gradient in the graph
 - Resistance is equal to $R = \frac{V}{I}$ or $\frac{1}{R} = \frac{I}{V}$
- As the resistance R decreases, the value of $\frac{1}{R}$ increases, so, the ratio $\frac{V}{I}$ increases
- So, current I increases with the p.d., with an increasing gradient since the resistance R is also decreasing as the p.d. increases
 - The graph that matches this description is option A

B is incorrect because the resistance of the thermistor decreases as p.d. increases, so it does not obey Ohm's Law

C is incorrect because the resistance of the thermistor decreases as p.d. increases, whereas this graph shows the resistance decreasing

D is incorrect because when the p.d. increases, the current increases so the graph should have a positive gradient

Exam Tip: It is important to note, that the gradient only tells us the way the resistance is changing (increasing or decreasing). We cannot use the value of the gradient to obtain the value of the resistance. Instead, we would need to consider the ratio of $\frac{V}{I}$ of a single point.

7

The correct answer is **C** because:

- Resistance R of a wire is equal to $R = \frac{\rho L}{A}$
- Cross-sectional area is equal to $A = \pi r^2 = \pi \left(\frac{d}{2}\right)^2 = \frac{\pi d^2}{4}$
- Cross-sectional area of one strand is equal to $= \frac{\pi (0.51 \times 10^{-3})^2}{4} = 2.0 \times 10^{-7} \text{ m}^2$
- Total cross-section of cable $= 12 \times (2.0 \times 10^{-7}) = 2.45 \times 10^{-6} \text{ m}^2$
 - $R = \frac{\rho L}{A} = \frac{(1.7 \times 10^{-8})(2000)}{(2.45 \times 10^{-6})} = 13.87 \Omega$

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 - $R = \frac{\rho L}{A} = \frac{(1.7 \times 10^{-8})(2000)}{(2.45 \times 10^{-6})} = 13.87 \Omega$
- Therefore, the resistance of the cable is **14 Ω**

8

The correct answer is **A** because:

- The resistance of an LDR decreases as the light intensity incident on it increases
- Ohm's law states or $V = IR$ or $R = \frac{V}{I}$
- The p. d. across a component is dependent on its resistance

- Since the resistance of the LDR decreases, the p. d. across it will also decrease
 - Therefore, if the original p. d. is 4 V, the only possible reading it can be when the light intensity is increased is 3 V

9

The correct answer is **D** because:

- Resistance R of a wire is equal to $R = \frac{\rho L}{A}$
 - $A = \frac{\rho L}{R} = \frac{(1.0 \times 10^{-6})(4.0)}{31} = 1.29 \times 10^{-7} \text{ m}^2$
- Cross-sectional area is equal to $A = \pi r^2$, so, $r = \sqrt{\frac{A}{\pi}}$
 - So, the radius is $= \sqrt{\frac{1.29 \times 10^{-7}}{\pi}} = 2.0 \times 10^{-4} \text{ m}$

10

The correct answer is **A** because:

- The resistance of a wire is given by $R = \frac{\rho L}{A}$
- If you stretch a wire to double its length, its cross-sectional area is halved
- This is because its volume must stay the same, i.e.:
 - If the volume of the unstretched wire is $V = AL$
 - Then, the volume of the stretched wire is $V = \left(\frac{A}{2}\right)(2L)$
- So, in the above formula, you are multiplying L by 2 and dividing A by 2, this means the resistance goes up by a factor of 4
 - $\frac{\rho(2L)}{\frac{1}{2}A} = 4\left(\frac{\rho L}{A}\right) = 4R$
- Therefore, the relationship between extension and resistance is a square law
 - $R \propto x^2$
- At zero extension, there will be some resistance, so, the graph does not start from the origin
 - Therefore, the graph that best fits this description is option A

B is incorrect because even though this graph shows a non-linear relationship, it shows an initial sharp rise in resistance which levels off and becomes constant, but this is not the case – the resistance will continue to rise with extension

C is incorrect because this graph shows the first section as a linear relationship which is incorrect

D is incorrect because this graph shows the resistance starts from the origin, but at zero extension, there will be some resistance so this is incorrect

